

Retail Store & Pricing Performance

Rossmann drugstores — 844,392 open-day records, 1,115 stores, 2013-01-01 to 2015-07-31

1,115	+38.8%	+21% / +14%	11
Stores analyzed	Promo sales uplift	Promo footfall vs basket	Underperformers flagged

So what: Promos pull in more shoppers far more than they grow basket size, and uplift is strongest Monday, weakest Friday. Meanwhile 11 stores have the foot traffic but not the sales — that's a pricing/conversion fix, not a marketing-spend fix, and it's a named, actionable list.

Key Findings

- Promo sales uplift is +38.8% overall, driven mostly by footfall (+21.2% customers) rather than basket size (+13.8% spend/customer).
- Promo uplift decays through the week: +57% Monday down to +22% Friday — early-week promo spend has more leverage than flat weekly spend.
- Store Type B is an outlier: ~2.5-3x the daily customers of other types but the lowest basket value — reads as a high-traffic convenience format, not a scaled-down version of the other types.
- Competition proximity has ~zero correlation with sales ($r = -0.042$); the 'very close to a competitor' bucket has the highest average sales of the four distance quartiles — likely a retail-cluster effect, not proof competition is harmless.
- 11 stores flagged with at-or-above-average footfall for their format but bottom-25% sales within that format — traffic isn't the problem, pricing/conversion likely is.

Charts

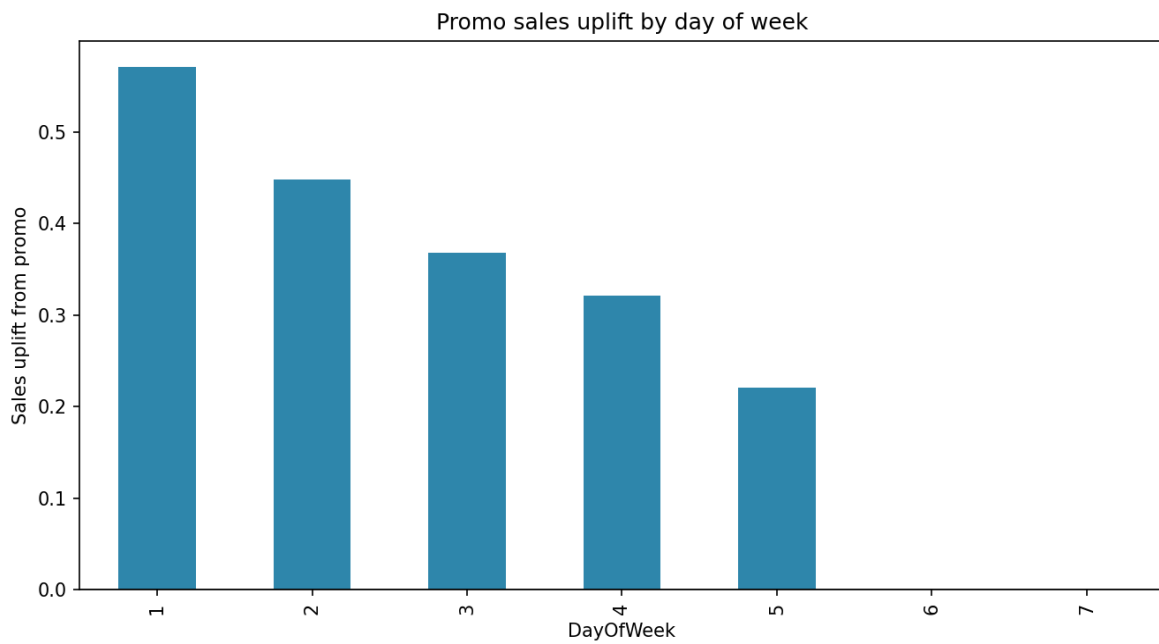


Fig 1. Promo sales uplift by day of week — decays from Monday to Friday.

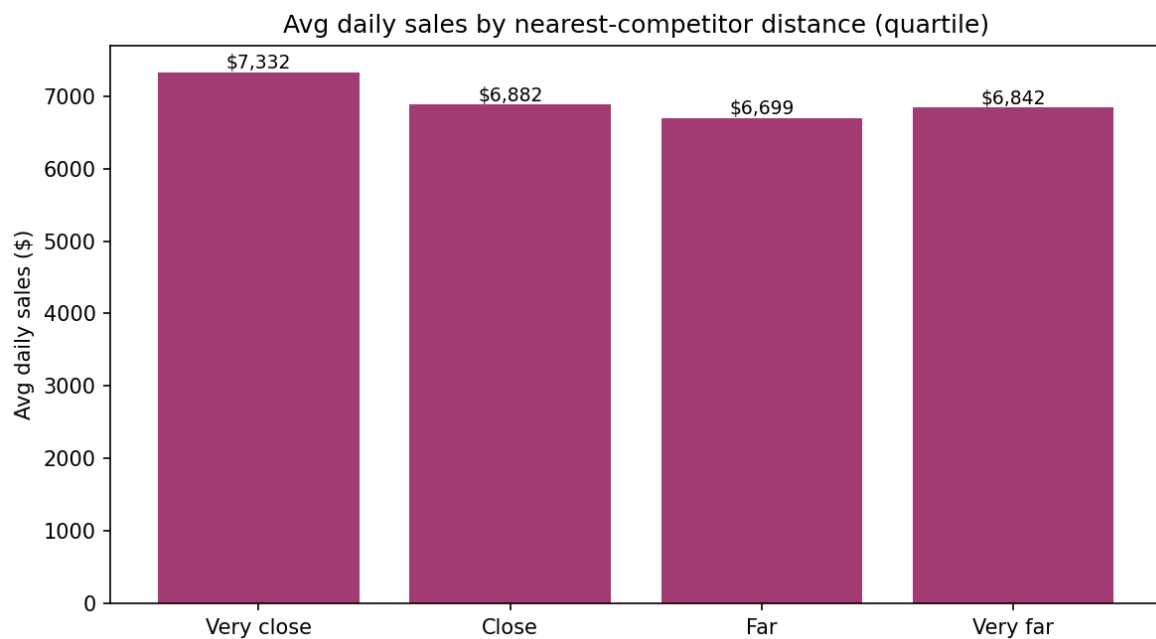


Fig 2. Avg daily sales by nearest-competitor distance — no meaningful penalty for proximity.

Method

- Filtered to Open == 1 (closed-day rows are zero by construction, not a performance signal).
- Promo effect controlled by day-of-week, since Rossmann promos only run Mon-Fri in this data.
- Competition distance bucketed into quartiles to avoid outlier-driven correlation reads.
- Underperformer flag = footfall percentile $\geq 50\%$ AND sales percentile $\leq 25\%$, both computed within store type to control for format differences.

Recommendation

- Shift promo timing/budget toward Monday-Tuesday where uplift is highest; Friday promos are the weakest lever in this data.
- Investigate the 11 flagged underperforming stores individually — look at local pricing, checkout staffing, or basket-building tactics before spending on more foot-traffic marketing there.
- Don't use competition distance as a site-selection signal off this data alone — the near-zero/slightly positive correlation likely reflects confounding by location quality, not a causal effect.

Source: Rossmann Store Sales (Kaggle). Full pipeline reproducible via [analysis.py](#); interactive version in [dashboard.html](#).